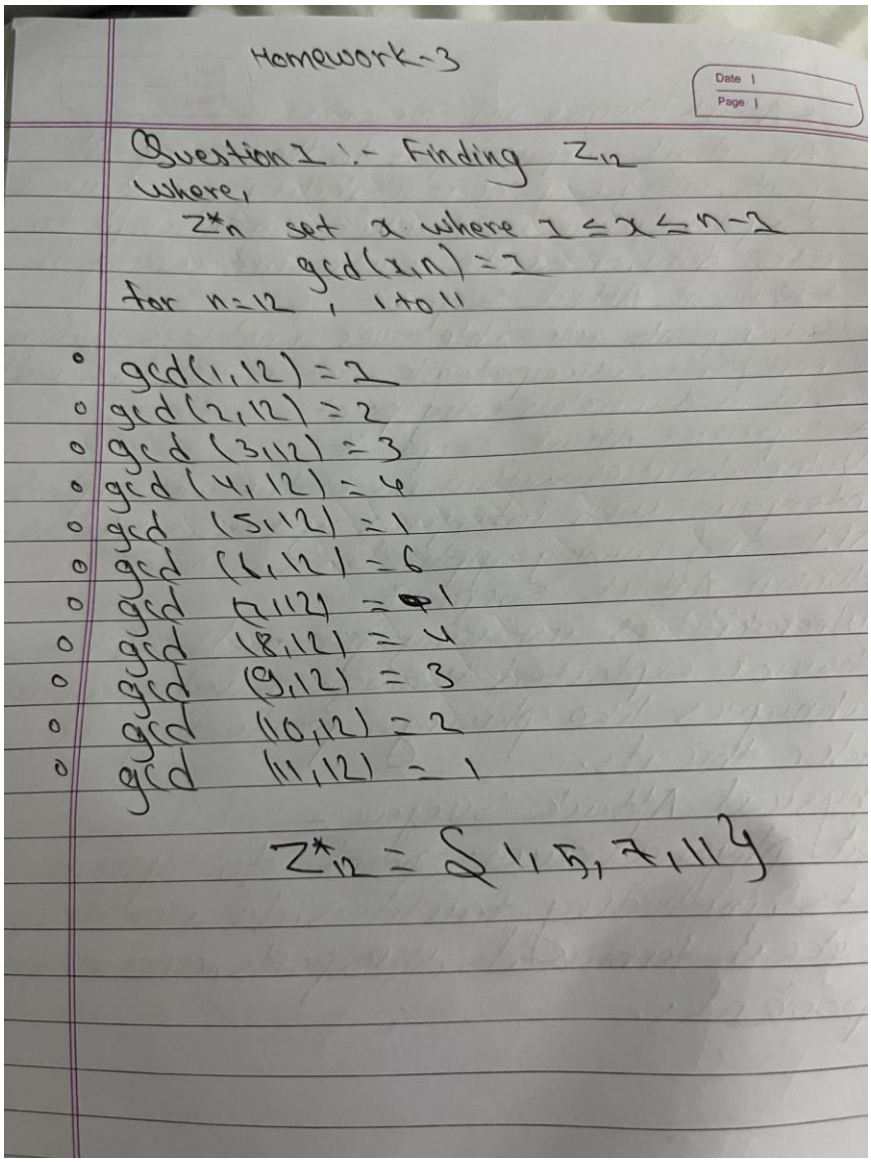
CEG 4424/6424 (Security Attacks and Defenses)
Homework #3

Student Name: _____Utsav Acharya_____
Wright UID: _____U01113272_____
Undergraduate/Graduate: ____UG_____

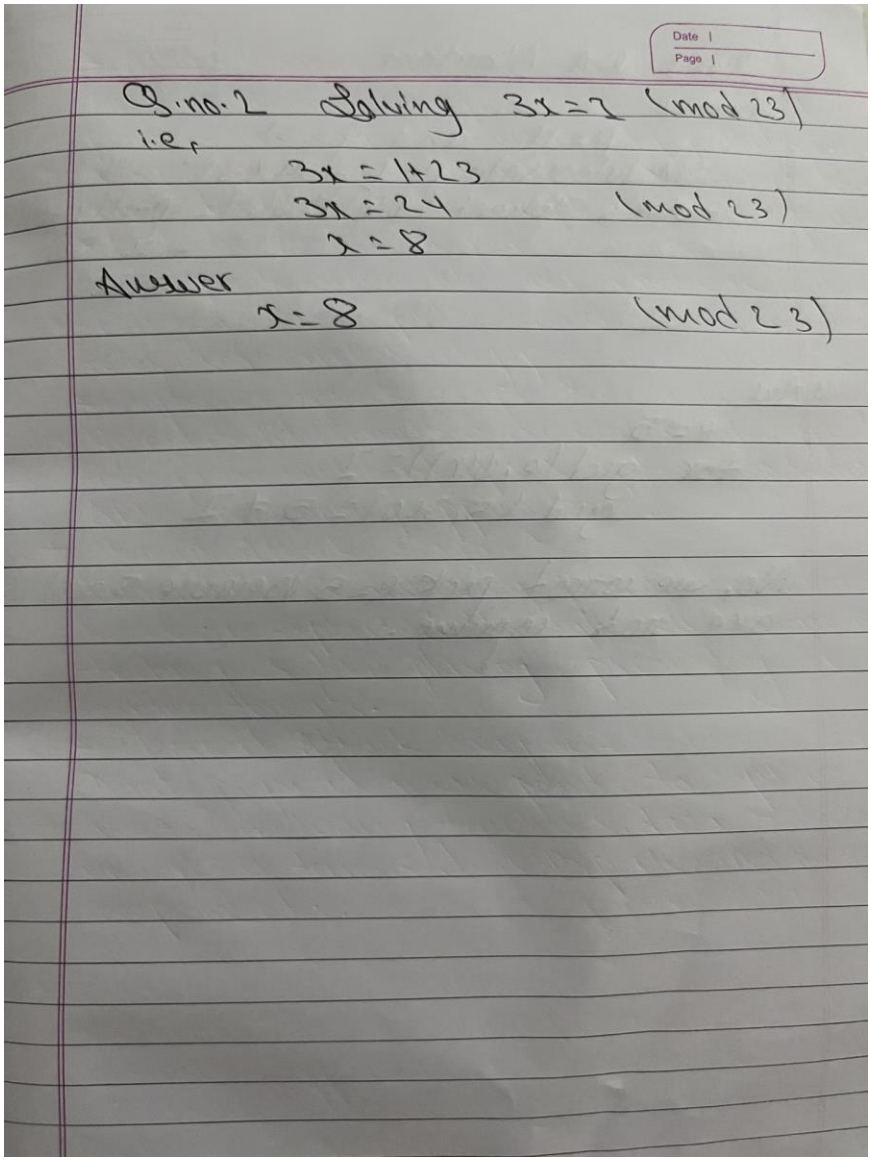
Question 1. (5 points) Let Zn* = {x| 1≤x ≤ n-1, where GCD(x, n)=1 }. Z12* = ?

Answer



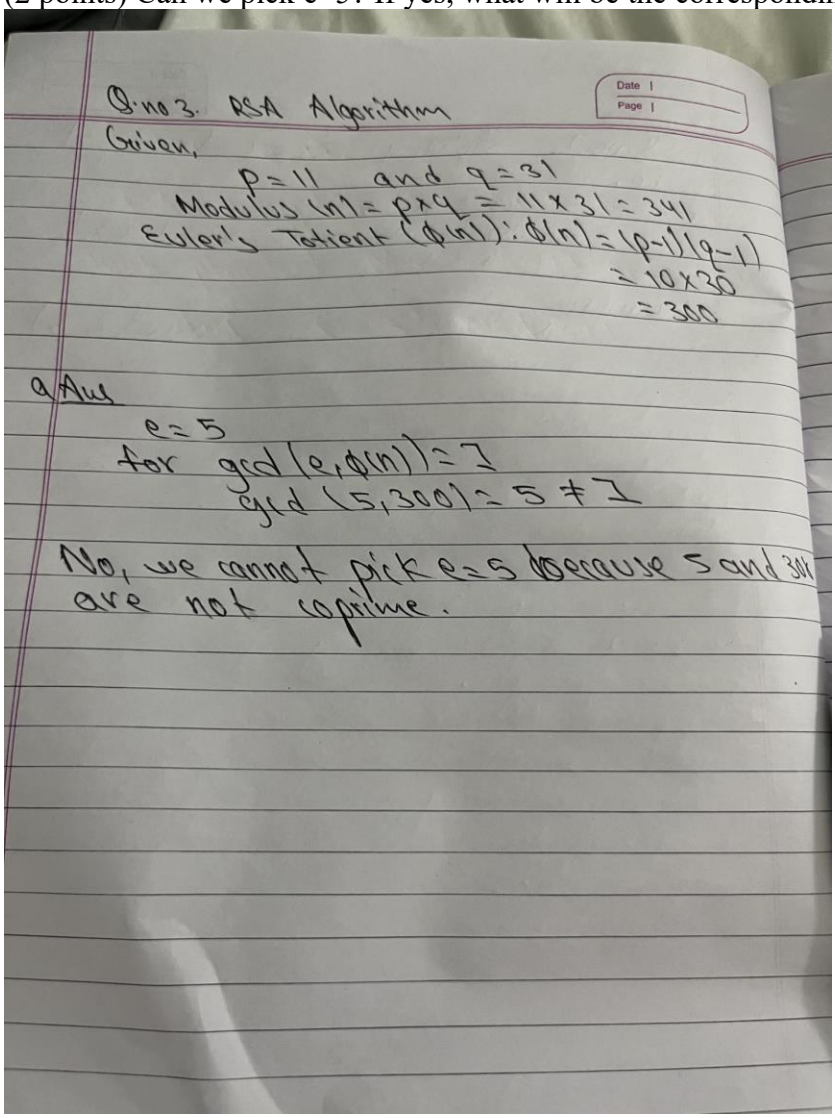
Question 2. (5 points) Find integer x such that 3*x = 1 (mod 23).

Answer



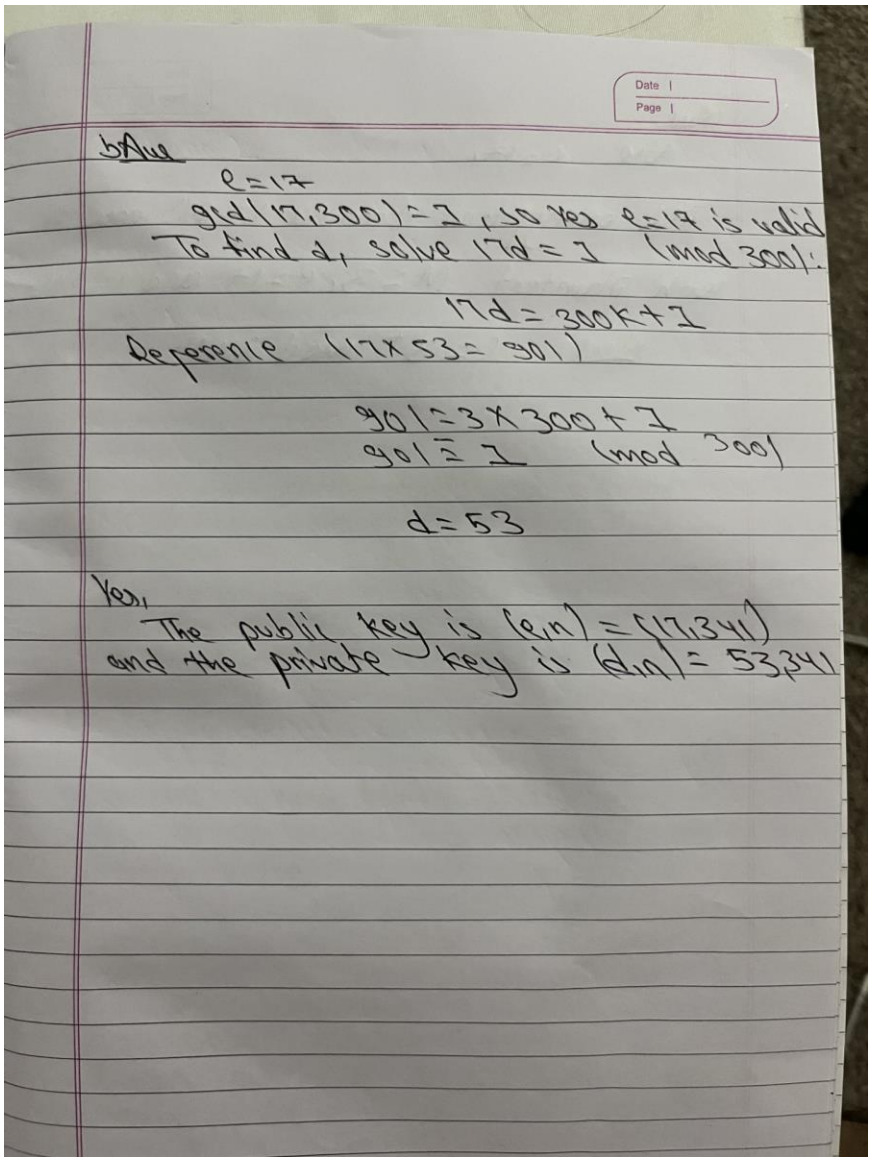
Question 3 (10 points). Consider the RSA algorithm. Let the two prime numbers, p=11 and q=31. You need to derive appropriate public key (e,n) and private key (d,n).

(a) (2 points) Can we pick e=5? If yes, what will be the corresponding (d,n)?



(b) (5 points) Can we pick e=17? If yes, what will be the corresponding (d,n)? Calculation Reference is given as follows.

17*51=867
17*52=884
17*53=901
17*54=918



(c) (3 points) Use e=17, how to encrypt the number 3? You do not need to provide the encrypted value.

Date
Page

Ans

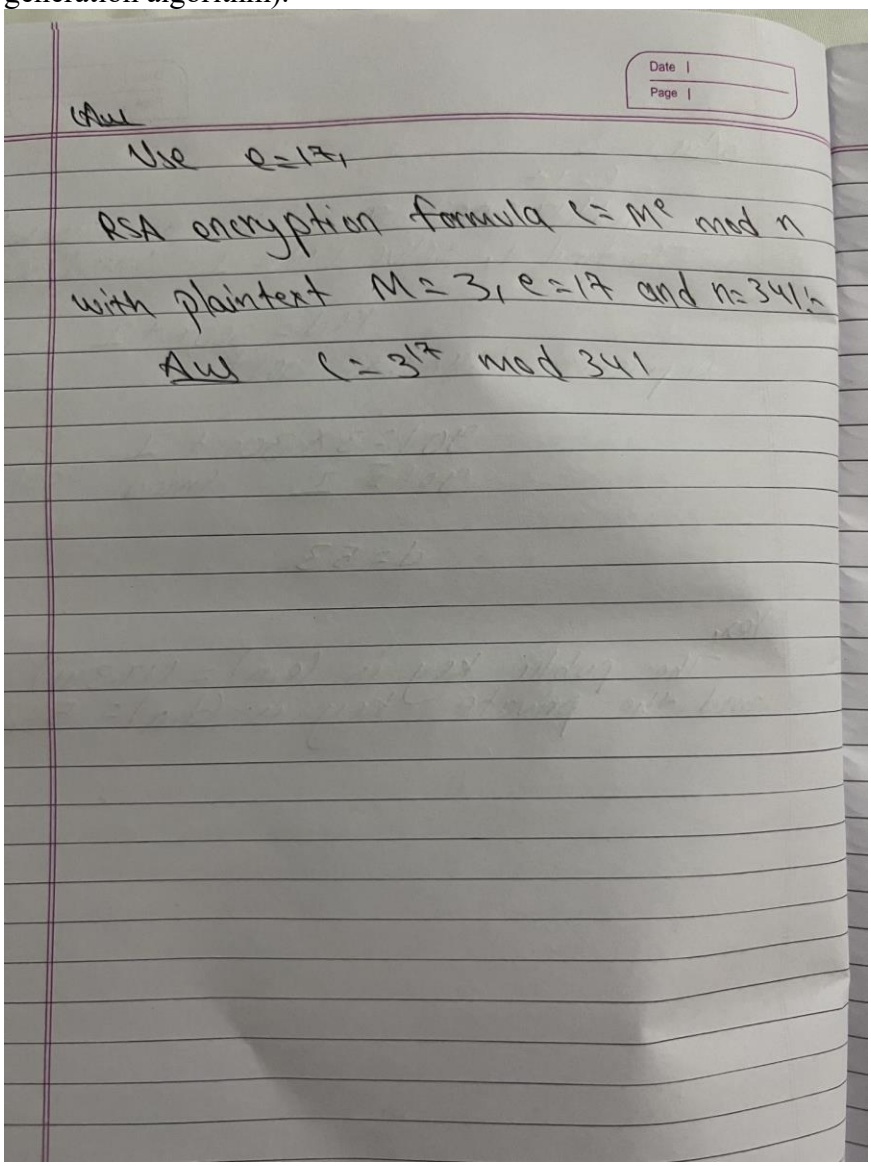
Use $e = 17$,

RSA encryption formula $C = M^e \bmod n$

with plaintext $M = 3$, $e = 17$ and $n = 341$

Ans $C = 3^{17} \bmod 341$

Question 4 (10 points, Question for Graduate Students) Is PU=(23, 55) and PR=(7,55) a valid RSA public key and private key pair? Show your work. (Hint: use the RSA key generation algorithm).



Question 5 (20 points). The Common Weakness Enumeration (CWE) is a category system for hardware and software weaknesses and vulnerabilities. Please describe what is CWE-322, CWE-328, CWE-336, and CWE-1240. (Hint: Please go to the website cwe.mitre.org to find the descriptions of these CWEs. You should describe the content in your own words by rephrasing the sentences from the online resources.**)**

Answer:

- **CWE-322 (Key Exchange without Entity Authentication):** Occurs when a system performs a cryptographic key exchange without verifying the identity of the participating parties. This leaves the channel open to Man-in-the-Middle (MitM) attacks where an adversary can intercept or modify communications.
- **CWE-328 (Use of Weak Hash):** Refers to relying on cryptographic hash algorithms that have known mathematical weaknesses or collision vulnerabilities (e.g., MD5 or SHA-1), allowing attackers to forge signatures or manipulate hashed data.
- **CWE-336 (Same Seed in PRNG):** Happens when a Pseudorandom Number Generator (PRNG) is repeatedly initialized using the exact same seed value, resulting in identical and predictable outputs across sessions.